

AdaViP: Task-Conditioned Visuotactile Perception via Hypernetworks for Contact-Rich Manipulation

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Abstract—Contact-rich manipulation requires a robot to change what it perceives and trusts as an interaction evolves. This paper studies task-conditioned adaptive visuotactile perception for contact-rich robotic manipulation.

I. INTRODUCTION

Contact-rich manipulation underpins many practical robotic skills, including insertion, peeling, wiping, tool use, and dexterous in-hand manipulation. These tasks cannot be solved reliably from vision alone: visual observations provide global geometry and free-space context, while tactile sensing reveals local contact, deformation, friction, and slip that may be occluded or too subtle to infer from images [1]–[4]. Proprioception further anchors these signals in the robot’s motion and force state. Effective manipulation therefore depends on combining heterogeneous sensory evidence throughout an interaction.

The central difficulty is that the value of each modality changes over time. During approach, visual geometry and object pose dominate. Near contact onset, small tactile transients and changes in end-effector state become decisive. During sustained contact, the policy must jointly interpret global visual context, local tactile deformation, and proprioceptive feedback. A perception stack with fixed parameters must nevertheless apply the same feature extraction strategy across all of these phases. This creates a structural mismatch between a changing physical interaction and a static sensory representation.

Existing visuotactile policies alleviate this mismatch mainly after features have already been computed. Concatenation and cross-attention combine modality embeddings; tactile gates suppress irrelevant touch; slow–fast controllers introduce high-rate tactile correction; and policy-consensus methods route among modality-specific action predictions [5]–[9]. Larger tactile-language and world-model systems similarly introduce contact-aware tokens, modality selection, or predictive objectives [10]–[16]. These mechanisms improve fusion and control, but they typically leave the upstream perception mapping fixed after training. As a result, the model can change how it weights a visual or tactile feature without changing what that feature extractor is optimized to represent at the current task stage.

Our core insight is that adaptive multimodal control should begin before policy inference. Rather than only deciding which modality to trust, a robot should also adjust what each modality extracts according to the task and the evolving interaction state. Conditional computation provides a suitable mechanism: hypernetworks can generate context-dependent parameters for another network [17], while related approaches such as dynamic filters and FiLM demonstrate that feature computation can be

conditioned without duplicating a complete policy [18], [19]. This suggests a compact alternative to maintaining separate perception-policy stacks for different contact regimes.

We introduce AdaViP, a task-conditioned visuotactile perception framework for contact-rich manipulation. AdaViP uses hypernetworks to adapt the multimodal perception pathway from task and execution context before the resulting representation is passed to a shared action policy. The formulation is intentionally centered on adaptive perception rather than policy selection: task conditioning changes the representation delivered by the visual, tactile, and proprioceptive streams, while the downstream policy remains shared across the rollout. This design targets heterogeneous sensory observations whose relevance changes sharply around contact, while avoiding a separate specialist policy for every task phase.

We evaluate AdaViP in both simulation and the real world. In simulation, we reformulate the nine ManiFeel tasks [20] as a joint multi-task benchmark and test AdaViP as a plug-and-play perception module across policy backbones. Targeted simulation ablations isolate the roles of task conditioning, execution progress, cross-modal fusion, and hypernetwork-based adaptation. Real-robot experiments then verify the same paired-backbone improvements on longer-horizon contact-rich manipulation under sensing noise, occlusion, and embodiment-specific dynamics.

The contributions of this work are:

- We formulate contact-rich manipulation as a problem of task-conditioned visuotactile perception, in which the sensory representation should change with the task and interaction state.
- We propose AdaViP, a hypernetwork-based framework that adapts heterogeneous visual, tactile, and proprioceptive perception before action prediction while retaining a shared policy backbone.
- We design a simulation-to-real evaluation that separates overall policy gains from targeted ablations of adaptive perception, task/progress conditioning, and cross-modal fusion.

II. RELATED WORK

A. Tactile Sensing and Representation Learning

High-resolution optical tactile sensors convert deformation of a compliant surface into image-like observations. GelSight established a widely used design for estimating local geometry and force [2]; DIGIT reduced the cost and form factor for in-hand manipulation [4]; GelSlim 3.0 jointly measures shape,

force, and slip in a compact finger [21]; and Soft-Bubble grippers combine dense visuotactile geometry with mechanical compliance [22]. Such sensors have enabled visuotactile tool use [23], tactile state estimation [24], and multimodal object tracking under occlusion [25].

A parallel line of work studies representations that are less tied to a single task or sensor. Self-supervised objectives align vision and touch for contact-rich manipulation [26], while contrastive pretraining improves multimodal transfer [27]. Sparsh learns general-purpose tactile encoders across several vision-based sensors [28], and AnyTouch unifies static and dynamic signals across multiple optical tactile platforms [29]. Other approaches explicitly model contact dynamics or physical state, including TactileVAD [30], Robot Synesthesia [31], and tactile pretraining for cable insertion [32]. These methods improve the information available to a policy; AdaViP instead studies how the mapping from sensory input to policy representation should change with task context and execution phase.

B. Visuotactile Policy Learning

Visuotactile policy learning spans grasp prediction, local feedback control, dexterous manipulation, and long-horizon imitation. Touch improves grasp-outcome prediction when combined with vision [3], supports local cloth control through visuotactile affordances [33], and enables in-hand manipulation when visual state estimates are ambiguous [34]. For dexterous imitation, Learning Visuotactile Skills with Two Multifingered Hands collects demonstrations with tactile-equipped hands [35], while 3D-ViTac fuses visual and tactile observations in a common 3D representation [36].

Sequence-policy backbones have accelerated this direction. Diffusion Policy models multimodal action distributions through conditional denoising [37], and ACT predicts temporally coherent action chunks [38]. RDP augments slow action generation with a fast tactile feedback path [5]; ReTac-ACT introduces cross-modal attention and state-conditioned tactile gating [6]; Seeing Touch from Motion models tactile temporal change and cross-modal interaction [7]; and Multi-Modal Policy Consensus routes among modality-specific diffusion policies [8]. GelFusion uses vision-led cross-attention to reduce visual degradation during fusion [9]. These methods primarily improve policy reactivity, tactile encoding, or downstream fusion. AdaViP addresses the complementary question of adapting the perception transformation that precedes the shared policy.

The same issue is emerging in larger multimodal policies. TLA and VTLa ground tactile sequences with language for contact-intensive insertion [10], [11]; FuSe adapts generalist robot policies to heterogeneous sensors through language grounding [12]; and TacVLA activates tactile tokens through contact-aware gating [13]. T-Rex scales tactile-reactive manipulation through variable-rate tactile processing [14], whereas OmniVTA and Dream-Tac introduce predictive visuotactile world or world-action models [15], [16]. These systems reinforce the need for selective multimodal computation, but

their scope differs from explicitly generating task- and phase-dependent perception transformations for a compact imitation-learning policy.

C. Task-Conditioned and Adaptive Perception

Conditional computation changes a model’s parameters or activations as a function of context. Hypernetworks generate weights for a target network [17]; Dynamic Filter Networks generate sample-specific convolutional filters [18]; FiLM modulates intermediate features with conditional affine transformations [19]; and Gated Multimodal Units learn multiplicative gates for information fusion [39]. These mechanisms provide different trade-offs between expressivity, computational cost, and architectural disruption.

In manipulation, task information is often concatenated with a task-agnostic observation representation or introduced only at the policy level. HyperTASR instead conditions visual representation transformations on both task specification and progression state [40]. AdaViP follows the broader premise that perception should adapt during execution, but addresses a different technical setting: heterogeneous vision, touch, and proprioception streams whose informativeness changes around contact. The goal is therefore not merely task-aware visual selection or scalar modality weighting, but context-dependent reconfiguration of the multimodal representation supplied to a shared control policy.

III. METHOD

A. Method Overview

Given synchronized visual observation o_v , tactile observation o_t , and proprioceptive state o_p , AdaViP first extracts base representations z_v^0 , z_t^0 , and z_p^0 with frozen modality-specific encoders. A set of conditioned HyperNet modules, shown compactly as one generator block in Fig. 1, receives the task embedding together with an execution-progress context derived from the current base representations. The HyperNets generate only the parameters of three modality-specific transformation encoders, while the corresponding decoders remain ordinary shared learned modules. This produces task- and phase-adaptive representations z_v , z_t , and z_p . A cross-modal transformer exchanges information among the transformed streams, after which a fourth generated fusion encoder and a shared fusion decoder form z_{fused} . The resulting representation is passed to an unchanged policy backbone, so AdaViP changes the upstream perception mapping rather than replacing or specializing the action policy.

IV. EXPERIMENTS

A. Experimental Setup

Simulation benchmark. We use the nine simulation tasks from ManiFeel [20], covering four insertion tasks (Peg Insertion, USB Insertion, Power Plug Insertion, and Gear Assembly), two screwing tasks (Nut-bolt Assembly and Bulb Installation), and three visually constrained exploration tasks (Peg Reorientation, Object Search, and Ball Sorting under dim lighting). For our study, we train a *single shared multi-task policy* across all nine

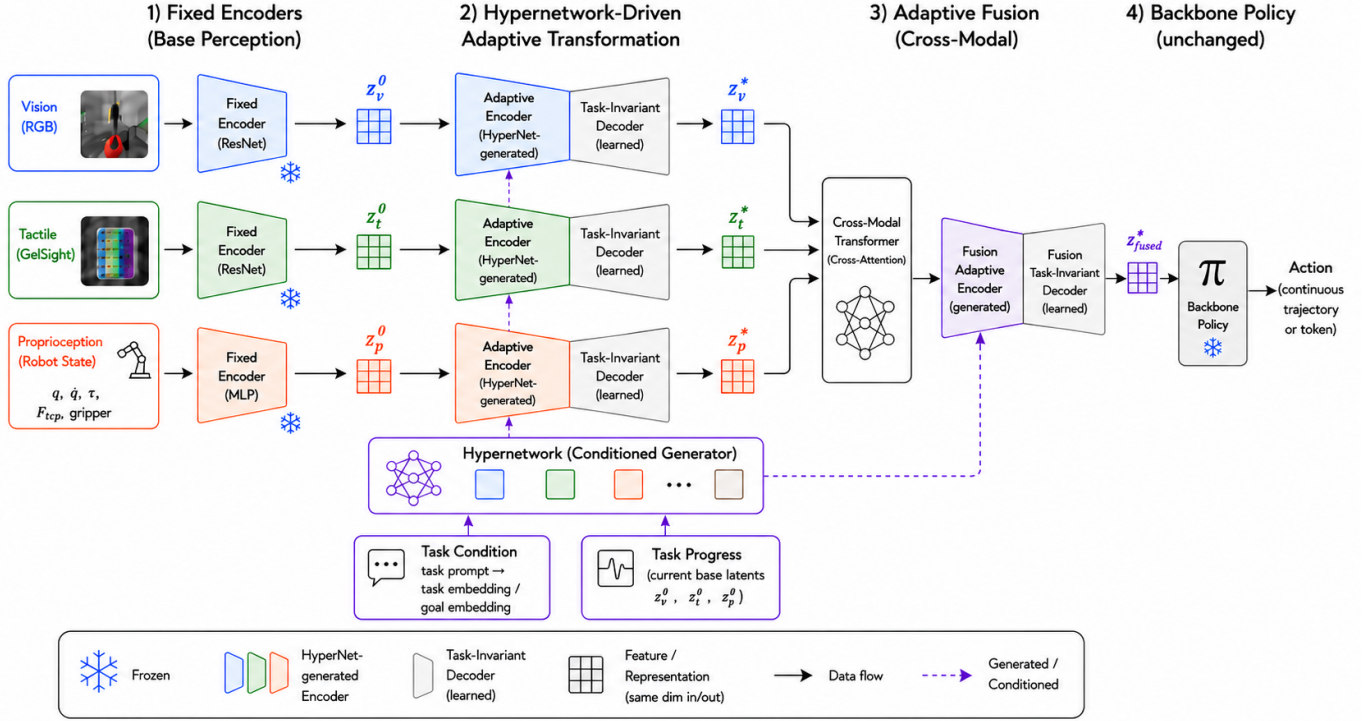


Fig. 1. Overview of AdaViP. Frozen modality encoders produce base visual, tactile, and proprioceptive representations. Task and execution context condition HyperNet modules that generate the transformation encoders for each modality and for fusion; the paired decoders remain shared and task-invariant. Cross-modal attention then integrates the transformed streams before an unchanged policy backbone predicts the action.

tasks. Each trajectory is paired with a task identifier that is embedded and supplied as the task condition to every multi-task method, while all methods share the same demonstrations, observation streams, action representation, task sampling, and train-validation split. Unless otherwise stated, visuotactile methods use the same visual, tactile, and proprioceptive inputs and the same fixed base modality encoders; the vision-only baseline removes only the tactile stream.

We evaluate two policy families: Diffusion Policy (DP) and a Flow-Matching policy. For each backbone, AdaViP is inserted only in the upstream perception pathway and the downstream action-policy architecture is otherwise unchanged. The primary simulation metric is task success rate (SR), reported per task and averaged across all nine tasks; results are reported as mean $\pm$ standard deviation over three random seeds. The ablation study uses the DP backbone and the identical multi-task protocol.

Real-robot evaluation. Real-world experiments use a physical single-arm platform with a parallel-jaw gripper, two RGB views (wrist-mounted and fixed third-person), GelSight optical tactile sensing at the gripper, and robot proprioception including end-effector pose and gripper state. We evaluate Peeling, Wiping, Occluded Object Picking, Threaded Assembly, and Puzzle Insertion, using shared teleoperated demonstrations and identical rollout protocols for all methods. All compared real-robot policies receive vision, touch, and proprioception; AdaViP is evaluated on both VT-DP and RDP backbones. The primary

metric is SR (%), reported per task and averaged across tasks.

TODO: The real-robot task set may be adjusted after final hardware integration.

B. Simulation Results

Table I evaluates the central claim in a controlled multi-task setting. The vision-only DP row provides a sensing sanity check, while paired comparisons between each visuotactile backbone and its AdaViP variant isolate the benefit of adaptive perception. Using two policy families tests whether AdaViP transfers across action-generation backbones rather than relying on a particular policy parameterization.

C. Real-World Results

Table II tests whether the paired improvements observed in simulation persist under real contact uncertainty, sensing noise, and longer-horizon execution. We keep the real-world comparison compact and use only complete visuotactile policies, since the purpose of this experiment is physical validation of AdaViP rather than re-benchmarking the general value of touch.

D. Ablation and Design Analysis

Table III concentrates the design ablations in simulation, where repeated evaluation is reproducible and inexpensive. Static AE tests whether a learned but task-invariant transformation is sufficient; FiLM replaces generated transformation

TABLE I

MULTI-TASK SIMULATION RESULTS ON MANIFEEL. SUCCESS RATE (SR, %) IS REPORTED FOR NINE TASKS UNDER ONE SHARED MULTI-TASK POLICY. ALL CONFIGURATIONS INCLUDE PROPRIOCEPTION. ADAViP IS EVALUATED AS A PLUG-AND-PLAY PERCEPTION MODULE ON TWO POLICY BACKBONES. VALUES ARE MEAN $\pm$ STD OVER THREE SEEDS.

Method	Input	Insertion				Screwing		Exploration			Avg.
		Peg Ins.	USB Ins.	Plug Ins.	Gear Asm.	Nut-bolt Asm.	Bulb Inst.	Peg Reorient.	Object Search	Ball Sort	
DP	V	–	–	–	–	–	–	–	–	–	–
DP	V+T	–	–	–	–	–	–	–	–	–	–
DP + AdaViP	V+T	–	–	–	–	–	–	–	–	–	–
Flow Matching	V+T	–	–	–	–	–	–	–	–	–	–
Flow Matching + AdaViP	V+T	–	–	–	–	–	–	–	–	–	–

TABLE II

MAIN REAL-ROBOT RESULTS. SUCCESS RATE (SR, %) IS EVALUATED USING THE SAME ROBOT, SENSOR SETUP, DEMONSTRATIONS, AND ROLLOUT PROTOCOL FOR ALL METHODS. ALL METHODS USE VISION, TOUCH, AND PROPRIOCEPTION.

Method	Peeling	Wiping	Occluded Picking	Threaded Assembly	Puzzle Insertion	Avg.
VT-DP	–	–	–	–	–	–
VT-DP + AdaViP	–	–	–	–	–	–
RDP [5]	–	–	–	–	–	–
RDP + AdaViP	–	–	–	–	–	–

TABLE III

ADAViP ABLATION ON MANIFEEL. ALL VARIANTS USE THE DP BACKBONE AND THE SAME MULTI-TASK PROTOCOL. CATEGORY AVERAGES SUMMARIZE FOUR INSERTION, TWO SCREWING, AND THREE EXPLORATION TASKS.

Variant	Insert.	Screw.	Explore.	Overall
Static AE	–	–	–	–
FiLM	–	–	–	–
AdaViP w/o Task Cond.	–	–	–	–
AdaViP w/o Progress	–	–	–	–
AdaViP w/o Cross-Attn	–	–	–	–
AdaViP (Full)	–	–	–	–

parameters with feature-wise affine modulation; the task- and progress-conditioning variants remove one source of context at a time; and removing cross-attention tests whether adaptive unimodal transformations alone explain the gain. All variants retain the complete visuotactile observation at the downstream policy input, so conditioning ablations change only the information available to AdaViP’s adaptive perception module.

V. CONCLUSION

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